

Composition Cluster B1.12 + B1.13 + B1.14 + B1.15 — The Co-Required Evolution Cluster Where Productive Tension Requires All Three Mechanisms, Mutation and Directed Selection Operate in Governance Opposition to Maintain Instinct/Reasoning Balance, and Action-Feedback Provides the Evidence Foundation for Evidence-Informed Directed Selection

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its contribution is to establish, as explicit public prior art, the co-required cluster relationship holding between B1.12 (productive tension as architectural property), B1.13 (instinct evolution through undirected mutation), B1.14 (DNA evolution through directed selection), and B1.15 (action-feedback evolution as the evidence-closing mechanism) — formalizing why these four commitments require each other’s presence, how they interact through three pairwise dependencies, and what architectural degradation follows from any partial-cluster configuration.

Abstract

The CKS theory Paper 2 (Li, April 2026) commits to three evolution mechanisms operating in productive tension under unified human governance: instinct evolution as undirected mutation (B1.13), DNA evolution as directed selection (B1.14), and action-feedback evolution as the evidence-closing loop (B1.15). Paper 2 further commits to productive tension among these three mechanisms as an architectural property in its own right (B1.12). This note formalizes the four commitments as a co-required cluster: B1.12 is not itself a mechanism but the integrating architectural property that emerges only when B1.13, B1.14, and B1.15 all operate together. Three pairwise dependencies hold within the cluster. B1.13 and B1.14 operate in governance opposition — mutation evolves the instinct layer while directed selection governs the reasoning layer, and each layer’s governance instruments are distinct, so changes to either layer create governance-attention pressure on the other, maintaining the instinct/reasoning separation of B1.01 through evolution rather than despite it. B1.14 and B1.15 form an evidence-governance pipeline — action-feedback produces evidence-based proposals; directed selection is the

authorized pathway through which proposals become DNA changes. B1.13 and B1.15 interact through mutation detection — behavioral shifts induced by LLM updates may surface as evidence patterns in the action layer, making action-feedback an instrument for post-deployment mutation impact assessment. The note catalogs four partial-cluster configurations, each with a distinct degradation profile, then specifies the operational integration the full cluster achieves and the detection tests by which the cluster's completeness can be verified. Cluster-level failure modes B3.13 (Single-Mechanism Evolution) and B3.24 (Evolution Stasis) are named and distinguished.

1. Why the cluster structure requires formalization

Phase B4 composition notes formalize cases where CKS architectural commitments are not merely adjacent but mutually constitutive: each commitment in the cluster requires the others' presence for its own architectural property to be achieved. The present note covers a four-commitment cluster: the most complex cluster structure Phase B4 has formalized to date.

The complexity warrants explicit treatment because the four commitments in this cluster span two distinct types of architectural object. B1.12 names an architectural property — productive tension among evolution mechanisms. B1.13, B1.14, and B1.15 name the three mechanisms whose joint operation produces that property. The relationship is not symmetric: B1.12 depends on all three mechanisms; the mechanisms do not depend on B1.12 for their own existence. But B1.12 is what justifies holding all three mechanisms together in one architecture — without the productive-tension framing, the three mechanisms could in principle be treated as independent options rather than as a required cluster. The cluster structure formalizes the entailment running in both directions: B1.12 requires all three mechanisms (operational dependency), and the three mechanisms together produce B1.12 (compositional dependency).

Within the three mechanisms, three pairwise dependencies hold that are distinct from the B1.12 integrating dependency. Each pairwise dependency names a specific interaction between two mechanisms that the other pairwise dependencies cannot substitute. Naming the pairwise dependencies individually is what makes the partial-cluster failure modes diagnosable: when one mechanism is absent, the specific interaction it participates in is what degrades, not a generic loss of “some evolution.”

This formalization is the public prior-art record for: (a) productive tension as a co-required architectural property rather than an optional design aspiration; (b) governance opposition between mutation and directed selection as a distinct architectural relationship; (c) the evidence-governance pipeline between action-feedback and directed selection as a specific compositional requirement; and (d) mutation detection through action-feedback evidence as an architectural interaction, not a monitoring afterthought.

2. Cluster identification: four commitments and the integrating architecture

Cluster members:

B1.12 — Productive tension as integrating commitment. Per Paper 2 §7.1 and §7.2, productive tension among the three evolution mechanisms is an architectural commitment: the architecture holds all three mechanisms simultaneously not as a coincidence of design but because each contributes what the others cannot, and their joint operation produces an evolutionary character no single mechanism achieves alone. B1.12 names this as a commitment the architecture makes, not a property it merely observes.

B1.13 — Instinct evolution as undirected mutation. Per Paper 2 §7.2, instinct evolution sharpens the LLM and substrate-platform infrastructure layer through upstream upgrades the Self does not author. Capability changes are mutation-like in arriving undirected — good or bad in any specific case — and are integrated through governed responses including verification substrates (B2.04 routing around bad instinct; B2.06 verification gates for behavioral compatibility). Governed through verification-substrate machinery per Paper 2 §8.2.

B1.14 — DNA evolution as directed selection. Per Paper 2 §7.2, DNA evolution operates on the CKS substrate's stabilized orchestration content under explicit goals defined by human governance. Selection criteria for what DNA changes are accepted, retained, or reverted are themselves substrate content — they are governable under the same authority architecture that governs the content they evaluate. Governed through the standard authority architecture per Paper 2 §8.2.

B1.15 — Action-feedback evolution as the evidence-closing mechanism. Per Paper 2 §7.2, action-feedback evolution accumulates lived operational experience recorded in the action layer (B2.26) and drives substrate content refinement and DNA refactoring. Proposals for DNA changes originate from evidence patterns accumulated in the action layer and pass through human mediation before becoming authorized substrate changes. Governed through human-mediated proposal-and-acceptance machinery per Paper 2 §8.2.

Cluster type: Co-required cluster with B1.12 as integrating commitment requiring the other three. The cluster is not a collection of related commitments that happen to coexist; it is a single architectural unit whose internal structure is specified by the integrating dependency and the three pairwise dependencies.

3. B1.12 as integrating commitment

B1.12 occupies a structurally distinct position within the cluster. It is not itself an evolution mechanism — it does not name a process that changes the system. It names the architectural property that emerges when all three mechanisms operate simultaneously on their respective layers under unified governance.

The source paper (§7.2) articulates what productive tension delivers that no single mechanism achieves alone: instinct evolution provides exploratory capacity — undirected capability jumps that discover possibilities directed selection would not search; DNA evolution provides directional capacity — purposeful change toward governance-defined goals with predictable trajectory; action-feedback evolution provides empirical feedback — operational experience grounding both

the directional change and the consumption of new instinct layers. The three contributions are non-substitutable: directional capacity does not replace exploratory capacity; empirical feedback does not replace either.

Because B1.12 is the property that emerges from the joint operation of B1.13, B1.14, and B1.15, it depends on all three architecturally. An architecture that declares “productive tension among evolution mechanisms” without instantiating all three mechanisms has named a goal without providing the machinery to achieve it. The declaration is architecturally vacuous without the full mechanism set. This is the integrating dependency: $B1.12 \leftarrow \{B1.13, B1.14, B1.15\}$.

The converse relationship also holds, at the cluster level: the three mechanisms together produce productive tension, and productive tension is what justifies holding all three mechanisms in the architecture rather than selecting the most convenient one. An architecture that has instinct evolution and DNA evolution but lacks action-feedback has two mechanisms without the empirical grounding that makes their interaction genuinely productive rather than merely concurrent. An architecture that has any two of the three without the third lacks one of the three non-substitutable contributions and thus does not achieve the productive tension B1.12 names.

4. Pairwise dependencies within the cluster

The three pairwise dependencies within the cluster are distinct relationships. Each captures a specific architectural interaction between two mechanisms; none substitutes for the others.

4.1 B1.13 + B1.14: governance opposition maintaining instinct/reasoning balance

Mutation (B1.13) and directed selection (B1.14) operate in governance opposition. The term names a precise architectural relationship: each mechanism evolves a different layer of the instinct/reasoning separation (B1.01), under a different governance instrument, with changes to one layer creating governance-attention pressure on the other.

Mutation operates on the instinct layer — it changes what the LLM does at the inference layer through upstream upgrades. Its governance instrument is verification-substrate machinery: parallel-run patterns, routing around bad instinct, behavioral compatibility testing, pinning of high-stakes decisions to the reasoning layer regardless of instinct capability. Directed selection operates on the reasoning layer — it changes DNA content (orchestration rules, selection criteria, reasoning substrate structure) through human-authorized modifications. Its governance instrument is the standard authority architecture from Paper 1: humans authorize DNA changes; LLMs do not commit DNA changes outside human authority.

The two governance instruments are distinct and non-interchangeable. Verification substrates cannot govern DNA changes; the authority architecture for DNA cannot govern LLM behavior at inference time. Because the instruments are distinct, each mechanism proceeds under its own governance discipline. But the mechanisms are not independent of each other’s effects. A mutation event — an LLM update that changes behavioral capabilities — may require verification against existing DNA rules: do the orchestration rules written for prior instinct behavior still hold for the new instinct behavior? If they do not, a directed selection event may be warranted.

Conversely, a directed selection event — a DNA change that modifies what the reasoning layer expects of instinct — may require mutation governance review: does instinct as currently deployed still satisfy the reasoning layer's new expectations?

The governance opposition is therefore productive in both directions. Mutation without directed selection means instinct evolves ungoverned while reasoning remains static — the instinct/reasoning balance drifts as instinct capability changes but reasoning governance does not adapt. Directed selection without mutation management means reasoning governance evolves but instinct changes proceed without behavioral compatibility verification. Both one-sided configurations break the instinct/reasoning balance over time. The governance opposition — both mechanisms operating under their respective instruments, with each triggering attention in the other's domain — is what maintains the instinct/reasoning separation of B1.01 through evolution rather than collapsing it.

4.2 B1.14 + B1.15: evidence-governance pipeline

Directed selection (B1.14) and action-feedback evolution (B1.15) form an evidence-governance pipeline. The pipeline has two stages that are architecturally distinct and jointly required.

Action-feedback evolution produces evidence-based improvement proposals. Operational experience accumulates in the action layer (B2.26) — recorded task instances, outputs, and their evidence patterns. Substrates that evaluate these patterns (Stage 1 evaluation, per Paper 2 §8.2) assess whether the accumulated evidence supports a proposal for DNA modification. Proposals name a specific DNA change and the evidence basis for it. This is the evidence foundation: proposals that originate in operational reality rather than in design judgment alone.

Directed selection is the governance mechanism through which proposals become authorized DNA changes. Human review of an action-feedback proposal is itself a directed selection event (Stage 2 authorization, per Paper 2 §8.2): a human exercises authority over what DNA change is accepted, retained, or rejected. The governance instrument for directed selection — human authority over substrate content per Paper 1's authority-not-labor commitment — applies at the authorization stage. The DNA change does not execute automatically; the proposal-and-acceptance machinery ensures that the evidence foundation informs rather than determines the governance decision.

The pipeline has two directions of dependency. Without directed selection as the authorization mechanism, action-feedback proposals have no governed pathway to DNA change: proposals accumulate but cannot become authorized substrate modifications, so the evidence loop is operationally inert. Without action-feedback as the evidence source, directed selection decisions are made without operational grounding: the governance is present but operates on judgment alone rather than on accumulated evidence from the system's own operation. The combination produces evidence-governed DNA evolution: proposals that are grounded in operational reality and authorized through human governance, rather than either automated drift or ungrounded judgment.

4.3 B1.13 + B1.15: mutation detection through operational evidence

Instinct evolution (B1.13) and action-feedback evolution (B1.15) interact through a third relationship: mutation behavioral changes may be detectable through action-feedback evidence patterns.

When an LLM update changes instinct behavior, the behavioral change manifests in the action layer (B2.26) as a pattern shift in how the system executes tasks. If the LLM now handles certain cases differently than the prior version handled them — producing different outputs, following different inference paths, failing cases the prior version handled — the action layer accumulates evidence of the behavioral shift. Substrates that evaluate action-layer evidence patterns for action-feedback proposals may detect mutation-induced behavioral changes as evidence that a DNA adjustment is warranted: for example, a verification rule written for prior instinct behavior may need revision for the new behavior, surfacing as a DNA change proposal.

This interaction is complementary to the governance opposition dependency of §4.1, but operates at a different point in the governance timeline. The governance opposition dependency handles the prospective governance question: when a mutation event is planned or integrated, does the reasoning layer governance need to adapt? The mutation-detection dependency handles the retrospective evidence question: after a mutation event has occurred and the system has operated under new instinct, does the action-layer evidence reveal behavioral shifts that warrant DNA review? Together, the two relationships provide governance coverage at both the integration stage and the post-deployment evidence stage.

The mutation-detection relationship also clarifies a directional asymmetry: instinct evolution (B1.13) is the upstream mechanism that produces behavioral changes; action-feedback (B1.15) is the downstream evidence layer that detects them. Mutation governance (verification gates per B2.06) provides pre-deployment governance; action-feedback evidence provides post-deployment impact assessment. Neither replaces the other. The two mechanisms together produce fuller mutation governance coverage than either alone.

5. Cluster co-presence requirement

The cluster co-presence requirement specifies that all four commitments — B1.12 and all three mechanisms — must be present for the cluster’s architectural properties to be achieved. The four partial-cluster configurations each produce a distinct degradation profile.

B1.12 without mechanisms (productive tension declared without machinery). An architecture that states productive tension as a design goal but instantiates only one or two mechanisms has named an aspiration without the operational structure to achieve it. No actual tension is produced; the declaration is architecturally vacuous.

B1.13 + B1.14 without B1.15 (mutation and directed selection without evidence). Evolution operates through instinct and reasoning governance, but without operational evidence informing directed selection decisions. Governance is judgment-only rather than evidence-informed. DNA changes reflect what governance decides is desirable without systematic grounding in what the

system's operation has revealed about actual performance patterns. The evidence-governance pipeline of §4.2 is absent; directed selection operates blind to the system's lived operational reality.

B1.14 + B1.15 without B1.13 (directed selection and action-feedback without mutation management). The DNA layer evolves through evidence-informed governance, and action-feedback proposals flow through the authorization pipeline. But instinct evolution is ungoverned: LLM updates proceed without verification-substrate machinery, without behavioral compatibility testing, and without the governance opposition that maintains instinct/reasoning balance during instinct changes. The instinct layer and the reasoning layer drift apart as instinct evolves without the directed selection review that governance opposition would trigger.

B1.13 + B1.15 without B1.14 (mutation and action-feedback without directed selection). Instinct evolves; operational evidence accumulates in the action layer; action-feedback substrates may identify behavioral patterns warranting DNA change. But no governed mechanism exists to translate proposals into authorized DNA modifications. Evidence accumulates without an authorized pathway to act on it. Proposals originate without a governance decision point. The evidence-governance pipeline of §4.2 is severed at the authorization stage: the evidence foundation exists but has nowhere to go.

Full cluster: all three mechanisms operational, B1.12 productive tension achieved. Instinct evolution proceeds under verification-substrate governance; DNA evolution proceeds under the authority architecture; action-feedback proposals flow through the evidence-governance pipeline. Mutation events trigger directed selection review; directed selection events prompt verification review of instinct assumptions; action-feedback evidence surfaces mutation-induced behavioral shifts. The evidence-mutation-governance pipeline is complete, B1.12 productive tension is achieved, and the instinct/reasoning separation of B1.01 is maintained through evolution rather than being eroded by it.

6. Operational integration

The complete cluster operates as a continuous governance cycle. The cycle does not have a canonical starting point; it is described here beginning with a mutation event for expository clarity.

An LLM vendor update arrives, triggering a mutation event per B1.13. Verification gates (B2.06) test behavioral compatibility: does the new LLM behavior satisfy existing orchestration rules? Does the instinct/reasoning boundary need adjustment? Routing per B2.04 manages deployment — the new instinct may be deployed under parallel-run verification, with the prior reasoning retained as fallback or as active verification on safety-critical paths. The mutation event is recorded as a governance event in the substrate.

The system operates under the new instinct configuration. The action layer (B2.26) accumulates operational evidence: task instances, outputs, evidence of behavioral patterns. Substrates evaluating action-layer evidence patterns (Stage 1 per Paper 2 §8.2) assess whether the

accumulated evidence supports DNA change proposals. Where the evidence supports a proposal — for example, where the new instinct behavior reveals that a prior verification rule is overly conservative, or where new behavioral patterns surface a governance gap — a proposal is formed and submitted for governance review.

Human governance reviews the proposal and authorizes, modifies, or rejects the proposed DNA change per B1.14 (Stage 2 per Paper 2 §8.2). This is a directed selection event: selection criteria are themselves substrate content and are applied under human authority. An authorized DNA change modifies the reasoning layer's orchestration rules or substrate structure. The DNA change may itself trigger a verification review of instinct assumptions — does the new reasoning layer configuration still fit the instinct layer as currently deployed? This triggers back into the mutation governance cycle, maintaining the governance opposition of §4.1.

The full cycle integrates all three mechanisms under unified governance: instinct evolution through upstream upgrade and verification machinery; DNA evolution through evidence-informed human authorization; action-feedback evidence accumulating continuously and feeding the directed selection pipeline. B1.12 productive tension is maintained not as a static property but as a continuously-operative governance dynamic.

7. Detection and cluster-level failure modes

Cluster completeness detection. Verifying that the full cluster is instantiated requires checking each mechanism individually and each pairwise relationship.

Individual mechanism checks: Is instinct evolution governed through verification-substrate machinery (B1.13 present)? Are verification gates configured for LLM update integration? Is DNA evolution governed through the authority architecture (B1.14 present)? Are selection criteria substrate content subject to human authority? Is action-feedback evolution instantiated with evidence accumulation in the action layer and a proposal pathway (B1.15 present)? Are substrates configured to evaluate action-layer evidence patterns and form proposals?

Pairwise completeness checks: For the B1.13+B1.14 governance opposition — do mutation events trigger directed selection review, and do directed selection events trigger instinct verification review? Is a process in place for verifying behavioral compatibility of DNA changes against current instinct behavior? For the B1.14+B1.15 evidence-governance pipeline — do action-feedback proposals produce directed selection events? Is the pathway from evidence pattern to authorized DNA change operationally active? For the B1.13+B1.15 mutation detection — does the action-layer evidence monitoring include detection of mutation-induced behavioral shifts?

Integrating check: Are all three mechanisms and their pairwise interactions operational, and does the system as a whole exhibit B1.12 productive tension — exploratory capacity from instinct evolution, directional capacity from directed selection, empirical feedback from action-feedback, all operating concurrently under unified governance?

Cluster-level failure modes. Two failure modes are named here at the cluster level; their full derivation is in separate notes.

B3.13 — Single-Mechanism Evolution covers partial cluster failures where one or two mechanisms are present but the full three-mechanism cluster is not. The degradation profile varies by which mechanism is absent, as §5 specifies. A system in a B3.13 failure mode may exhibit functioning evolution through the present mechanisms while missing the specific architectural interactions that absent mechanisms would contribute. Detection: the pairwise completeness checks above identify which interactions are missing.

B3.24 — Evolution Stasis covers complete cluster failure where no evolution mechanism is operationally instantiated. The system's architecture names evolution commitments that are not operational — verification-substrate machinery is not configured, DNA changes are not authorized through a governed process, action-feedback evidence accumulates without a proposal pathway. The system is architecturally static even if its components individually change. Detection: none of the individual mechanism checks in the detection section above are satisfied.

The two failure modes are distinct in kind. B3.13 is a partial-cluster failure producing specific degraded evolutionary dynamics; B3.24 is a complete absence of evolutionary dynamics. An architecture diagnosed with B3.24 has not yet instantiated the cluster; an architecture diagnosed with B3.13 has instantiated part of the cluster and requires the missing mechanism or interaction to achieve the full cluster's architectural properties.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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